AI RESEARCH TECHNOLOGY PLATFORM

AudioSurvey AI

Technical Architecture & System Documentation

Comprehensive reference for system design, APIs, data models, and operations

DOCUMENT VERSION	**1.2.0**
RELEASE DATE	**April 6, 2026**
AUTHOR	**Krishnanand**
PROJECT STATUS	**Completed**
CLASSIFICATION	**Confidential — Restricted Distribution**
CORE STACK	**Python · Flask · Twilio · Azure Cognitive Services · OpenAI Whisper · PyTorch**

ABSTRACT

AudioSurvey AI is an AI-powered multilingual Interactive Voice Response (IVR) survey platform designed for academic and public health field research. The system conducts fully automated telephone surveys in Kiswahili, targeting African refugee populations, through a combination of Twilio cloud telephony, Azure Neural Text-to-Speech, OpenAI Whisper speech recognition, and a four-stage ML post-processing pipeline comprising noise removal, transcription, translation, and English audio generation. Survey data is exported to structured Excel workbooks for analysis. The platform is managed via a secure web-based admin dashboard with real-time participant tracking, scheduled outbound calling, inbound call handling, and conference call capabilities.

Project Documentation

Revision History

Version	Date	Author	Change Description
0.1.0	2026-01-22	Krishnanand	Initial project setup — TTS, transcription, and translation modules scaffolded
0.2.0	2026-01-22	Krishnanand	IVR pipeline operational — inbound and outbound calling working end-to-end
0.3.0	2026-01-22	Krishnanand	Security improvements — `.env` for secrets, safe config template added
0.4.0	2026-01-23	Krishnanand	README added, contacts CSV excluded from version control
0.5.0	2026-01-26	Krishnanand	State management, call logs, and caller handler functions updated
0.6.0	2026-02-22	Krishnanand	5-March calling milestone — initial full call flow verified
1.0.0	2026-02-22	Krishnanand	**First stable release** — core IVR engine, Twilio integration, Whisper STT, admin dashboard
1.0.1	2026-02-26	Krishnanand	Excel export support for survey responses
1.0.2	2026-03-01	Krishnanand	Excel formatting corrections, UX/UI upgrade, audio channel fix for full-call recordings
1.0.3	2026-03-03	Krishnanand	Updated survey questions, system cleanup
1.0.4	2026-03-04	Krishnanand	English translation export, "Convert to English" button added to dashboard
1.0.5	2026-03-05	Krishnanand	Authentication system — login/logout, PBKDF2 password hashing, brute-force protection, session management
1.0.6	2026-03-05	Krishnanand	Full call testing milestone — results saved and verified end-to-end
1.0.7	2026-03-08	Krishnanand	MCQO digit bug fix, hardcoded button logic corrected
1.0.8	2026-03-10	Krishnanand	DeepFilterNet background noise removal integrated, random crash fix, log improvements
1.0.9	2026-03-15	Krishnanand	macOS DMG packaging, icon generator update
1.1.0	2026-03-17	Krishnanand	File naming convention refactored for clarity; auth info cleanup; DOCUMENTATION.md created

Version	Date	Author	Change Description
1.1.1	2026-03-21	Krishnanand	Full audio recording logic redesigned — recording now starts at `/voice` entry, not after survey completion
1.1.2	2026-04-01	Krishnanand	Study guide and presentation documentation added (`STUDY_GUIDE.md`)
1.1.3	2026-04-05	Krishnanand	Inbound call response collection — inbound callers now matched to participants and routed through full survey + export flow
1.1.4	2026-04-05	Krishnanand	Call direction tracking — `direction` field added to call log and dashboard display (`Incoming` / `Outgoing`)
1.2.0	2026-04-06	Krishnanand	**Major release** — improved inbound call handling, auto participant creation for unknown callers, per-call End Call controls, End All Calls, live tech info panel, `runtime_status.py` module, redesigned login page, safer call/export state updates

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1. Executive Summary

1.1 Purpose

AudioSurvey AI is an AI-powered multilingual Interactive Voice Response (IVR) survey platform designed to conduct automated voice-based research surveys over telephone calls. The system targets **African refugees who speak Kiswahili**, enabling researchers to collect structured survey responses at scale without requiring in-person enumerators.

1.2 Business Context

The platform was built for academic and public health research. It is designed as a **flexible, multi-survey system** — the survey topic, questions, and thematic sections are fully configurable via the `data/questions.txt` file. Researchers can deploy different surveys for different studies without any code changes. Both outbound scheduled calling and inbound participant-initiated calls are fully supported, routing through the same survey and ML processing pipeline.

1.3 Key Capabilities

Capability	Description
Automated Outbound Calling	Scheduled batch dialing of research participants via Twilio
Inbound Call Support	Participants can call in; auto-matched to existing records or auto-created
Call Direction Tracking	All calls tagged as `inbound` or `outbound-api`, shown in dashboard
Multilingual IVR	Survey prompts spoken in Kiswahili via Azure Neural TTS
Multi-Format Questions	INFO, OPEN (speech), MCQ (DTMF keypad), MCQO (MCQ + "Other" speech)
Full-Call Recording	Complete audio capture of every call (up to 30 minutes)
AI Audio Processing	Background noise removal via DeepFilterNet
Speech-to-Text	Post-call transcription using OpenAI Whisper (large-v3)
Machine Translation	Automatic Kiswahili to English translation
English Audio Generation	TTS synthesis of translated responses
Structured Data Export	Excel export of MCQ/MCQO responses (original + English)
Admin Dashboard	Real-time web UI — participant management, live status, call controls
Per-Call & Global Call Controls	End individual calls or all active calls from the admin UI
Conference Calling	Three-way call support for researcher-moderated interviews
Runtime Status Panel	Live tech info — scheduler status, worker status, system load
macOS Distribution	Packaged as a native .app inside a DMG installer

1.4 Technology Summary

Layer	Technology
Runtime	Python 3.x
Web Framework	Flask 3.1.2
Telephony	Twilio Voice API
Live Speech Recognition	Twilio `<Gather>` (Kiswahili `sw-KE`)
Post-Call Transcription	OpenAI Whisper `large-v3`
Text-to-Speech (Prompts)	Azure Cognitive Services (Neural SSML)
Text-to-Speech (Output)	Google TTS (gTTS)
Translation	Google Translate (googletrans)

Layer	Technology
Noise Reduction	DeepFilterNet (PyTorch-based)
Audio Processing	FFmpeg, pydub
Data Export	pandas + openpyxl
Tunneling	ngrok
Runtime Monitoring	`runtime_status.py` (heartbeat-based)
Packaging	macOS .app bundle + DMG

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2. System Architecture

2.1 High-Level Architecture

```
+-----------------------------------------------------------------------+
|                         EXTERNAL SERVICES                             |
|   +------------------+  +------------------+  +------------------+    |
|   | Twilio Voice API |  | Azure Cognitive  |  | Google Translate |    |
|   |                  |  | Services (TTS)   |  |                  |    |
|   +--------+---------+  +--------+---------+  +---------+--------+    |
|            |                     |                      |             |
+-----------------------------------------------------------------------+
             |                     |                      |
             v                     v                      v
+-----------------------------------------------------------------------+
|                    FLASK APPLICATION SERVER (:5050)                    |
|                                                                       |
|   +--------------+  +--------------+  +--------------+                |
|   | Auth Module  |  | IVR Webhook  |  |    Admin     |                |
|   |              |  | Handlers     |  |  Dashboard   |                |
|   +--------------+  +--------------+  +--------------+                |
|                                                                       |
|   +--------------+  +--------------+  +--------------+                |
|   | Conference   |  | Excel Export |  | Inbound Call |                |
|   | Call Module  |  |              |  | Handler      |                |
|   +--------------+  +--------------+  +--------------+                |
|                                                                       |
|   +-----------------------------+  +-----------------------------+    |
|   |     BACKGROUND SERVICES     |  |   RUNTIME STATUS MODULE     |    |
|   |  +----------+ +-----------+ |  |  Scheduler & Worker         |    |
|   |  | Scheduler| | ML Worker | |  |  heartbeat tracking         |    |
|   |  | (15s)    | | (5s poll) | |  |                             |    |
|   |  +----------+ +-----------+ |  +-----------------------------+    |
|   +-----------------------------+                                     |
+-----------------------------------------------------------------------+
             |                     |
             v                     v
+-----------------------------------------------------------------------+
|                      ML PROCESSING PIPELINE                           |
|   +--------------+  +-----------+  +------------+  +-----------+      |
|   | DeepFilterNet|->| Whisper   |->| Translation|->| English   |      |
|   | (Denoise)    |  | large-v3  |  | (sw -> en) |  | TTS       |      |
|   +--------------+  +-----------+  +------------+  +-----------+      |
+-----------------------------------------------------------------------+
             |
             v
+-----------------------------------------------------------------------+
|                       LOCAL FILE STORAGE                              |
```

```
|                                                                        |
|   data/state/          data/audio/           data/audio_processed/    |
|   data/transcripts/    data/translations/    data/english_audio/      |
|   data/results/        data/ivr_audio/                                |
+------------------------------------------------------------------------+
```

Data flow summary:

- Participant Phone (outbound or inbound) <--> Twilio <--> ngrok <--> IVR Webhook Handlers
- Inbound callers matched to participants by phone or auto-created
- Admin Browser --> ngrok --> Auth Module --> Dashboard
- IVR Handlers --> Azure TTS --> data/ivr_audio/
- IVR Handlers --> Recording Done --> ML Worker
- ML Worker --> DeepFilterNet --> Whisper --> Translation --> English TTS
- Translation --> Google Translate API
- Scheduler --> Twilio (dial eligible participants)
- Excel Export --> data/state/ --> data/results/

2.2 Network Topology

```
+-------------------------------------------------------------------+
|                          INTERNET                                 |
|                                                                   |
|   +------------------+          +------------------+              |
|   | Participant      |          | Admin Browser    |              |
|   | (Kiswahili       |          |                  |              |
|   |  Speaker)        |          +--------+---------+              |
|   +--------+---------+                   |                        |
|            |                             |                        |
|   PSTN/VoIP (outbound or inbound)        | HTTPS                  |
|            v                            v                         |
|   +------------------+                                            |
|   | Twilio Cloud     |                                            |
|   | (Voice Infra)    |                                            |
|   +--------+---------+                                            |
|            |                                                      |
+-------------------------------------------------------------------+
             | HTTPS Webhooks
             v
+-------------------------------------------------------------------+
|                    LOCAL MACHINE (macOS)                           |
|                                                                   |
|   +------------------+         +------------------+               |
|   | ngrok Process    | <-----> | Flask Server     |               |
|   | :4040 (API)      | local   | :5050            |               |
|   +------------------+ host    +--------+---------+               |
|                                         |                         |
|                                         | REST API                |
|                                        v                          |
|                                +------------------+               |
|                                | Azure TTS        |               |
|                                | (East US Region) |               |
|                                +------------------+               |
+-------------------------------------------------------------------+
```

2.3 Thread Architecture

```
+------------------------------------------------------------------+
|                    PYTHON MAIN PROCESS                            |
|                                                                  |
|   +--------------------+                                         |
|   | Main Thread        |---> start_background_services()         |
|   | (Flask WSGI)       |          |            |                 |
|   +--------------------+          |            |                 |
|                                   v            v                 |
|   +--------------------+   +--------------------+                |
|   | Scheduler Thread   |   | Worker Thread      |                |
|   | (daemon, 15s loop)|   | (daemon, 5s poll) |                |
|   +--------------------+   +--------------------+                |
|            |                       |                             |
|    mark_scheduler_heartbeat()  mark_worker_heartbeat()           |
|            |                       |                             |
|            +----------+------------+                             |
|                       |                                          |
|                       v                                          |
|              +------------------+                                |
|              | threading.RLock |                                 |
|              | (STATE_IO_LOCK) |                                 |
|              +--------+--------+                                 |
|                       |                                          |
|                       v                                          |
|              +------------------+                                |
|              | participants.json                                 |
|              | call_log.csv                                      |
|              | settings.json   |                                 |
|              +------------------+                                |
+------------------------------------------------------------------+
```

---

3. Component Design

3.1 Module Dependency Graph

```
run_app.py ---------(subprocess)---------> twilio_handler.py
(Launcher)                                   (Flask App + IVR)
                                                    |
          +----------+----------+---------+---------+---------+------+
          |          |          |         |         |         |      |
          v          v          v         v         v         v      v
     dashboard.py  scheduler.py  state.py  utils.py  export_excel.py
     (Admin UI)    (Call Sched.) (State)   (Helpers) (Excel Export)
          |          |                                      |
          v          v                                      v
       state.py   state.py                            translate.py
     runtime_status.py

twilio_handler.py ---+----------+-----------+
                     |          |           |
                     v          v           v
            background_worker.py  transcribe.py  file_naming.py
            (ML Pipeline)         (Whisper STT)  (Safe Names)
                     |
         +------------+----------+-----------+
         |            |          |           |
         v            v          v           v
audio_preprocess.py  transcribe.py  translate.py  tts.py
(DeepFilterNet)      (Whisper)      (Translation) (Google TTS)
```

```
main.py -------> transcribe.py, translate.py, tts.py
(Batch Pipeline)

Supporting:  logger.py (Colored Logging)
             auth.py (Authentication)
             azure_tts.py (Azure TTS)
             runtime_warnings.py (Warning Suppression)
             twilio_utils.py (Twilio Call Helpers)
             runtime_status.py (Heartbeat Monitoring)  [NEW v1.2.0]
```

3.2 Module Descriptions

Module	Lines	Responsibility
`twilio_handler.py`	~1,780	Core Flask application. IVR webhook routes, inbound call matching, Azure TTS, auth, conference calling, recording download, call controls, Excel export trigger, application bootstrap. Central orchestrator.
`dashboard.py`	~1,280	Admin web dashboard. HTML UI with inline CSS/JS. Participant management, live-polling, scheduling, pause/resume, state reset, call controls (End Call, End All), live tech status panel.
`runtime_status.py`	~92	Background service health tracking. Heartbeat-based scheduler and worker status reporting. Exposes `get_runtime_snapshot()` for dashboard live info.
`state.py`	~241	Thread-safe participant state management. JSON persistence with atomic writes, call eligibility logic (`can_call`), state transitions, retry gap enforcement, participant schema migration.
`export_excel.py`	~314	Builds structured Excel exports from participant responses. Decodes DTMF digits back to option text, filters OPEN responses, supports original-language and English-translated exports.
`background_worker.py`	~148	Continuous polling worker that processes completed recordings through the 4-stage ML pipeline (denoise → transcribe → translate → TTS). Terminal progress bars and heartbeat updates.
`scheduler.py`	~114	Timed call dispatcher. Runs every 15 seconds, checks participant eligibility, places Twilio calls. Supports force-dial mode. Updates scheduler heartbeat.
`audio_preprocess.py`	~210	Audio noise removal pipeline using DeepFilterNet. FFmpeg-based channel extraction/resampling, PyTorch noise removal, output resampling for Whisper.
`transcribe.py`	~48	Whisper large-v3 speech-to-text. Swahili language hint, returns text and detected language. Single-file and directory batch modes.
`translate.py`	~134	Chunked Google Translate integration. Splits on sentence boundaries, retries failed chunks, marks failures with placeholder tags.
`tts.py`	~55	Google TTS (gTTS) wrapper for generating English MP3 audio from translated text.
`auth.py`	~165	Authentication helpers. Brute-force protection, session management, credential verification.
`utils.py`	~25	Scheduling utility. Converts NYC local time to UTC and updates participant state.
`azure_tts.py`	~49	Standalone Azure TTS module with SHA1 disk caching. Generates IVR call prompt audio.
`file_naming.py`	~28	Safe filename generation. Sanitizes participant IDs, generates timestamped base names.

Module	Lines	Responsibility
`logger.py`	~76	Colored console logging via `colorlog`. Silences noisy third-party library logs. Context manager for quiet operations.
`runtime_warnings.py`	~16	Suppresses urllib3/OpenSSL compatibility warnings at startup.
`run_app.py`	~129	Application launcher. Auto-starts ngrok, sets environment variables, launches Flask subprocess, opens browser.
`main.py`	~28	Standalone batch processor for offline audio → transcript → translation → TTS pipeline.

4. Data Flow & Pipelines

4.1 Outbound Call Lifecycle

```
ADMIN                    DASHBOARD           SCHEDULER           TWILIO
  |                          |                   |                  |
  |-- Upload CSV ---------->|                   |                  |
  |                          |-- upsert_participant()              |
  |                          |                   |                  |
  |-- Set schedule -------->|                   |                  |
  |                          |-- schedule_participant()            |
  |                          |                   |                  |
  |-- Click "Start" ------->|                   |                  |
  |                          |-- set_paused(false)                 |
  |                          |                   |                  |
  |                          |    +--[Every 15 seconds]--+         |
  |                          |    | mark_scheduler_heartbeat()     |
  |                          |    | load_participants()  |         |
  |                          |    | can_call() check     |         |
  |                          |    +----------------------+         |
  |                          |                   |                  |
  |                          |                   |-- calls.create --->
  |                          |                   |-- mark_call_started()
  |                          |                   |                  |

TWILIO               PARTICIPANT          IVR HANDLER        AZURE TTS
  |                          |                   |                  |
  |-- Ring ---------------->|                   |                  |
  |<-- Answer -------------|                   |                  |
  |-- POST /voice ------------------------->|   direction=outbound|
  |                          |              | Start recording     |
  |                          |              |-- Redirect /start   |
  |-- POST /start ------------------------->|                     |
  |                          |              |-- Generate TTS --->|
  |                          |              |<-- MP3 audio ------|
  |<-- TwiML: Play intro + Gather -------|                     |
  |                          |                   |                  |
  |    +--[For each question]--+                |                  |
  |    | Speech / DTMF input   |                |                  |
  |    | POST /next?q=N        |                |                  |
  |    | Store response        |                |                  |
  |    | Generate TTS          |                |                  |
  |    +-----------------------+                |                  |
  |                          |                   |                  |
  |<-- Play "Kwaheri" + Hangup ------------|                     |
  |-- POST /call-status (completed) ------>|                     |
  |                          |              | mark_call_result()  |
  |                          |              | append_to_excel()   |
```

```
     |-- POST /recording-done --------------->|                 |
     |                      |               | Download WAV       |
     |                      |               | processing = pending
     |                      |
```

4.2 Inbound Call Lifecycle (v1.2.0)

```
   PARTICIPANT           TWILIO            IVR HANDLER        STATE
     |                      |                   |                 |
     |-- Dials Twilio #---->|                   |                 |
     |                      |-- POST /voice --->|                 |
     |                      |   From=+254...    |                 |
     |                      |   Direction=inbound                 |
     |                      |                   |                 |
     |                      |        Match caller by phone?       |
     |                      |              |          |           |
     |                      |             Yes         No          |
     |                      |              |          |           |
     |                      |         Use existing   Auto-create  |
     |                      |         participant    new participant
     |                      |         record         from caller # |
     |                      |              |          |           |
     |                      |              +----+-----+           |
     |                      |                   |                 |
     |                      |              mark_call_started()    |
     |                      |              (direction=inbound)    |
     |                      |                   |                 |
     |                      | [Same IVR flow as outbound]         |
     |                      |   /start, /next, /mcq-handler, etc. |
     |                      |                   |                 |
     |                      |-- POST /call-status               |
     |                      |-- POST /recording-done            |
     |                      |              append_to_excel()     |
     |                      |              processing = pending  |
```

4.3 ML Processing Pipeline

```
  INPUT         STAGE 1              STAGE 2       STAGE 3          STAGE 4     OUTPUTS
            Audio Preprocessing   Transcription   Translation      English TTS

 +--------+  +----------------+    +------------+  +--------------+  +----------+  +------------
+
 | Raw    |  | FFmpeg         |    |            |  | Language     |  |          |  | Cleaned
|
 | WAV    |->| Channel Mix +  |    |            |  | Detection    |  |          |  | WAV
|
 |        |  | Resample 48kHz|    |            |  |   |          |  |          |  +------------
+
 +--------+  +-------+-------+    |            |  |lang=en lang |  |          |  |
Transcript|
                   |              |            |  |  |    !=en  |  |          |  | .txt (sw)
|
           +-------+-------+    | Whisper    |  |  v       v    |  |          |  +------------
+
           | DeepFilterNet |    | large-v3   |  | Copy  Chunk |  |          |  |
Translation
           | Noise Removal |    | lang="sw" |->| as-is  3000 |->| gTTS     |  | .txt (en)
|
           | (PyTorch)     |    | temp=0.0   |  | (en)  chars |  | lang=en |  +------------
```

```
+
          +-------+-------+    | fp16=False|  |  |    |     |  |         |  | English
|
                  |            +-----------+  |  |  Translate  |         |  | .mp3
|
          +-------+-------+                 |  |  3 retries  +---------+  +-----------
+
          | FFmpeg        |                 |  +------+------+
          | Resample 16kHz|                +---------|------+
          +---------------+                          v
                                              data/translations/
```

4.4 Recording Callback Flow

```
  Twilio POST /recording-done
       |
       v
  Recording completed?
       |           |
      No          Yes
       |           |
  Return 200    Recording URL present?
                   |           |
                  No          Yes
                   |           |
              Return 400    Find participant by CallSid
                               |
                        Known participant?
                           |           |
                          No          Yes
                           |           |
                  Download WAV    Call status retryable failure?
                  (saved)         (no-answer/busy/failed/canceled)
                                      |           |
                                     Yes         No
                                      |           |
                                 Skip pipeline  Participant engaged?
                                                  |           |
                                                 No          Yes
                                                  |           |
                                            Skip pipeline   Download WAV from Twilio
                                            (saved_no_engagement)  |
                                                          Save to data/audio/
                                                               |
                                                          Set processing_status =
pending
                                                               |
                                                          Log to call_log.csv
                                                               |
                                                          Background Worker picks up
```

4.5 Excel Export Data Flow

```
  INPUT                          PROCESSING                          OUTPUT

  +---------------+     +----------------------+     +---------------------------+
  | questions.txt |---->| Build Response       |     | ivr_responses.xlsx        |
  +---------------+     | Metadata             |     | (Kiswahili option text)   |
```

```
                      +----------+----------+     +-------------+-------------+
+-----------------+              |                              |
| participants.json|-->| Filter Responses   |     Cell-by-cell translate
+-----------------+   | (exclude OPEN)     |                    |
                      +----------+----------+     +-------------+-------------+
                                 |                | Translation Cache (dict)  |
                      +----------+----------+     +-------------+-------------+
                      | Decode DTMF         |                   |
                      | digit -> Option Text|     +-------------+-------------+
                      +----------+----------+     | ivr_responses_english.xlsx|
                                 |                +---------------------------+
                      +----------+----------+
                      | Renumber Question   |
                      | Keys (q1, q2, ...)  |
                      +----------+----------+
                                 |
                      +----------+----------+
                      | Build DataFrame     |-------> ivr_responses.xlsx
                      +---------------------+
```

---

5. API Reference

5.1 Public Endpoints (No Auth Required)

These endpoints are accessible without authentication — called by Twilio webhooks or used for health checks.

`POST /voice`

> **Call Entrypoint — Outbound and Inbound**

Initiates full-call recording and redirects to the survey start. For inbound calls, attempts to match the caller by phone number or auto-creates a participant.

Parameter	Source	Description
`CallSid`	Twilio	Unique call identifier
`From`	Twilio	Caller phone number (used for inbound matching)
`Direction`	Twilio	`inbound` or `outbound-api`

Response: TwiML — starts `<Recording>`, redirects to `/start`

`POST /start`

> **Survey Introduction**

Plays the first 2 intro prompts (INFO questions), then begins the first survey question.

Parameter	Source	Description
`CallSid`	Twilio	Call identifier

Response: TwiML — plays intros, gathers first OPEN question

`POST /next?q={index}`

> **Survey Question Router**

Core question loop. Stores previous answer, advances to next question based on type.

Parameter	Source	Description
`q`	Query string	0-based question index
`CallSid`	Twilio	Call identifier
`SpeechResult`	Twilio	Transcribed speech (OPEN questions)

Response: TwiML — varies by question type (Gather speech / Gather DTMF / Play + Redirect)

`POST /mcq-handler?q={index}`

> **MCQ/MCQO DTMF Handler**

Processes keypad digit. If MCQO and "Other" selected, redirects to speech capture.

Parameter	Source	Description
`q`	Query string	Question index
`Digits`	Twilio	Pressed DTMF digit (1–9)
`CallSid`	Twilio	Call identifier

Response: TwiML — next question or `/mcqo-other-handler`

`POST /mcqo-other-handler?q={index}`

> **MCQO "Other" Speech Handler**

Captures the free-speech "Other" response and moves to next question.

Parameter	Source	Description
`q`	Query string	Question index
`SpeechResult`	Twilio	Free-speech "Other" response

Response: TwiML — redirects to `/next?q={q+1}`

`POST /call-status`

> **Twilio Call Status Webhook**

Receives call completion events. Updates participant state, triggers Excel export on completion.

Parameter	Source	Description
`CallSid`	Twilio	Call identifier
`CallStatus`	Twilio	`completed`, `no-answer`, `busy`, `failed`, `canceled`

Response: `200 OK`

`POST /recording-done`

> **Twilio Recording Completion Webhook**

Downloads the completed recording WAV and queues it for ML processing.

Parameter	Source	Description
`CallSid`	Twilio	Call identifier
`RecordingUrl`	Twilio	URL to download recording
`RecordingStatus`	Twilio	`completed` or other
`Direction`	Twilio	`inbound` or `outbound-api`

Response: `200 OK`

`GET /health`

> **Health Check**

Response: `200 OK` — body: `ok`

`POST /conference_host`, `POST /conference_join`, `POST /conference_ivr`, `POST /conference_ivr_next`

> **Conference Call Endpoints**

Three-way conference call flow. Host hears IVR questions while waiting; conference starts when questions complete.

`POST /silence`

> **Silence Generator**

Returns TwiML with a 60-second pause — used as conference hold fallback.

`GET /ivr-audio/<filename>`

> **Static IVR Audio Server**

Serves cached Azure TTS audio from `data/ivr_audio/`.

5.2 Authenticated Endpoints (Login Required)

All `/admin/*` routes require an active session or valid `ADMIN_TOKEN` query parameter.

`GET /admin` — Admin Dashboard (full HTML page)

`GET /admin/live_state` — Live State JSON API

```
{
  "total": 42,
  "counts": {"pending": 10, "in_progress": 5, "completed": 25, "failed": 2},
  "participants": [
    {
      "participant_id": "P001",
      "phone_masked": "+2******1234",
      "status": "completed",
      "attempts": 1,
      "engaged": true,
      "direction": "inbound",
```

```
      "scheduled_local": "2026-04-05 14:00",
      "processing_status": "completed"
    }
  ],
  "runtime": {
    "scheduler": {"status": "running", "label": "Running", "age_sec": 3},
    "worker": {"status": "running", "label": "Running", "age_sec": 2}
  }
}
```

`POST /admin/upload_contacts` — **Upload participant CSV (`participant_id,phone_e164`)**

`POST /admin/save_questions` — **Save `data/questions.txt`**

`POST /admin/schedule` — **Set scheduled call time for a participant (NYC timezone)**

`POST /admin/dial_now` — **Force dial all eligible participants immediately**

`POST /admin/pause` **/** `POST /admin/resume` — **Toggle scheduler pause**

`POST /admin/reset_state` — **Back up and reset all participant state**

`GET /admin/export_excel` — **Download `ivr_responses.xlsx` (original language)**

`GET /admin/export_excel_english` — **Download `ivr_responses_english.xlsx` (translated)**

`POST /admin/conference_call` — **Dial two numbers into a conference room**

`POST /admin/end_call` — **End a specific active call by CallSid *(v1.2.0)***

`POST /admin/end_all_calls` — **End all currently active calls *(v1.2.0)***

5.3 Authentication Endpoints

`GET /login` — **Login page**

`POST /login` — **Authenticate; sets session cookie on success**

`POST /logout` — **Clear session and log event**

---

6. Data Models & State Schema

6.1 Participant Schema

```
{
  "P001": {
    "status": "completed",
    "attempts": 1,
    "last_call_time": "2026-04-05T18:30:00",
    "last_call_sid": "CA1234567890abcdef",
    "last_call_status": "completed",
    "engaged": true,
    "last_recording_url": "https://api.twilio.com/2010-04-01/Accounts/.../Recordings/RE...",
    "last_outputs": {},
    "scheduled_time_local": "2026-04-05 14:30",
```

```
      "scheduled_time_utc": "2026-04-05T18:30:00Z",
      "phone_e164": "+254700000000",
      "responses": {
        "q1": "Jina langu ni Amina",
        "q2": "2",
        "q3": "1",
        "survey_q_counter": 15
      },
      "processing_status": "completed",
      "audio_path": "data/audio/P001_20260405_183000.wav",
      "recording_url": "https://api.twilio.com/...",
      "direction": "inbound"
    }
  }
```

6.2 Participant State Machine

```
                         +-------+
                         | START |
                         +---+---+
                             |
                         CSV upload / inbound call
                             |
                             v
                         +-------+
                  +----->| idle  |
                  |      +---+---+
                  |          |
                  |     Schedule set / inbound triggers
                  |          |
                  |          v
                  |    +----------+      Eligible when:
                  +----|          |      - scheduled_time_utc <= now
                  |    | pending  |      - retry gap (1h) elapsed
                  |    |          |      - attempts < 3
                  |    +----+-----+
                  |         |
                  |    Call placed
                  |         |
                  |         v
                  |  +-------------+
                  |  | in_progress |
                  |  +------+------+
                  |         |
             +----+---------+---------+-------------------+
             |              |                             |
       No answer /    Survey finished             Completed but
       Busy           (engaged=true)              not engaged
       (attempts<3)         |                             |
             |              v                             |
             |       +-----------+                   (retry)
             +       | completed |
                     +-----------+

       No answer/Busy (attempts >= 3)
             |
             v
        +--------+
        | failed |
        +--------+
```

6.3 Processing Status State Machine

```
Recording received
       |
       v
Engaged?
 |         |
No        Yes
 |         |
saved_no_  pending
engagement  |
           Worker picks up
            |
            v
        processing
            |
       +----+----+
       |         |
    success    error
       |         |
  completed   failed
```

6.4 Call Log Schema

`data/state/call_log.csv`:

Column	Type	Description
`timestamp_utc`	ISO 8601	Event timestamp
`participant_id`	String	Participant identifier
`phone_masked`	String	Masked phone (e.g., `+2******1234`)
`direction`	String	`inbound` or `outbound-api`
`call_sid`	String	Twilio call SID
`recording_url`	URL	Twilio recording URL
`audio_path`	String	Local WAV file path
`transcript_path`	String	Kiswahili transcript path
`translation_path`	String	English translation path
`english_audio_path`	String	English MP3 path

6.5 Auth State Schema

`data/auth_state.json`:

```
{
  "fails": {
    "username|ip_address": [1711234567, 1711234580]
  },
  "locks": {
    "username|ip_address": 1711235467
  }
}
```

6.6 Settings Schema

`data/state/settings.json`:

```
{ "paused": false }
```

---

7. Authentication & Security

7.1 Authentication Flow

```
  USER                    FLASK              config.yaml     auth_state.json
auth_log.jsonl
   |                        |                     |                |                |
   |-- GET /admin --------->|                     |                |                |
   |                        |-- before_request    |                |                |
   |<-- 302 -> /login ------|   guard             |                |                |
   |                        |                     |                |                |
   |-- POST /login -------->|                     |                |                |
   |  (username,password)   |                     |                |                |
   |                        |-- _is_locked? ------|--------------->|                |
   |                        |                     |                |                |
   |           [If Locked]  |                     |                |                |
   |<-- "Try again in Xs"---|                     |                |                |
   |                        |                     |                |                |
   |                        |-- Load auth.users-->|                |                |
   |                        |-- check_password_hash()              |                |
   |                        |                     |                |                |
   |           [If Invalid] |-- _record_fail() --|--------------->|                |
   |                        |   (7 fails=lock)    |                |                |
   |<-- "Invalid creds"     |-- Log event --------|----------------|--------------->|
   |                        |                     |                |                |
   |           [If Valid]   |-- _clear_fails() --|--------------->|                |
   |                        |-- Set session       |                |                |
   |                        |-- Log success ------|----------------|--------------->|
   |<-- 302 -> /admin ------|                     |                |                |
```

7.2 Security Measures

Measure	Implementation	Details
Password Hashing	PBKDF2-SHA256	1,000,000 iterations via Werkzeug
Brute-Force Protection	Rate limiting per username+IP	7 failures in 10 min → 15-min lockout
Session Security	Flask sessions	`HttpOnly`, `SameSite=Lax`, `Secure=True`, 8-hour lifetime
Phone Masking	`mask_phone()`	First 2 + last 4 digits only (e.g., `+2******1234`)
PII Protection	`.gitignore`	Contacts CSV, participant state, audio, and logs excluded from git
Route Protection	`@app.before_request`	All `/admin` routes require session or admin token
Webhook Passthrough	Allowlist	Twilio webhook paths bypass auth

Measure	Implementation	Details
Audit Logging	JSONL	Every login/logout logged with IP, user-agent, timestamps
Atomic Writes	`os.replace()`	Prevents corrupt state files on crash

8. Survey Question Engine

8.1 Question File Format

```
TYPE|Question Text|Option1|Option2|Option3...
```

8.2 Supported Question Types

Type	Format	Input Method	Response Storage	Example
INFO	`INFO|text`	None	Not stored	`INFO|Maswali sehemu ya kwanza`
OPEN	`OPEN|text`	Speech (`<Gather input="speech">`)	Raw speech text	`OPEN|Tafadhali sema jina lako`
MCQ	`MCQ|text|opt1|opt2|opt3`	DTMF keypad	Digit (1–9)	`MCQ|Nani..?|Marafiki|Mume|Watoto`
MCQO	`MCQO|text|opt1|opt2|Nyingine`	DTMF + optional speech	Digit (1–9)	`MCQO|Kupanga..?|Ndiyo|Hapana|Nyingine`

8.3 IVR Question Flow

```
Call Answered (inbound or outbound)
     |
     v
/voice: Start Full-Call Recording (30 min max), tag direction
     |
     v
/start: Play Q[0] + Q[1] (INFO intros)
     |
     v
/start: Play Q[2] (First real question) with <Gather>
     |
     v
+--- Question Type? ---+------------------+-------------------+
|                       |                  |                   |
INFO                   OPEN               MCQ               MCQO
|                       |                  |                   |
Play text              Play text          Play text          Play text
Pause 1s               Gather speech      + options          + options
Redirect next          timeout=6s         Gather 1 digit     Gather 1 digit
                        |                  |                   |
                       Store              Store           Digit == Other?
                       SpeechResult       digit             |          |
                       mark_engaged()     mark_engaged()  No          Yes
                        |                  |                |          |
                        |                  |           Store      "Umechagua
                        |                  |           digit      nyingine..."
```

```
                         |                 |              |   Gather speech 4s
                         |                 |              |          |
    +---+-----------------++-----------------+--------+------+----------+
        |
        v
    Advance to Q[n+1]
        |
    More questions? Yes -> [loop]
                    No  -> "Kwaheri" + Hangup
```

8.4 TTS Prompt Generation

- **Voice:** `sw-KE-ZuriNeural` (Swahili) / `en-US-JennyNeural` (English)
- **Rate:** `-15%` prosody for clearer, slower delivery
- **Caching:** SHA1 hash of `voice|format|text` → disk-cached MP3 in `data/ivr_audio/`
- **Serving:** Public URL via `/ivr-audio/{hash}.mp3`, referenced in TwiML `<Play>` tags
- **SSML:** Full XML with entity escaping for special characters

8.5 MCQ Option Verbalization

```
"finya 1 kwa Ndiyo. finya 2 kwa Hapana. finya 3 kwa Nyingine."
```

("press 1 for Yes. press 2 for No. press 3 for Other.")

---

9. Configuration Reference

9.1 Environment Variables (`.env`)

Variable	Required	Default	Description
`TWILIO_ACCOUNT_SID`	Yes	—	Twilio account SID
`TWILIO_AUTH_TOKEN`	Yes	—	Twilio auth token
`TWILIO_FROM_NUMBER`	Yes	—	Twilio phone number (E.164)
`PUBLIC_BASE_URL`	Yes	—	Public HTTPS URL for webhooks
`ADMIN_TOKEN`	No	`""`	Optional token for admin access without login
`AZURE_SPEECH_KEY`	Yes	—	Azure Cognitive Services subscription key
`AZURE_SPEECH_REGION`	Yes	—	Azure region (e.g., `eastus`)
`AZURE_TTS_VOICE_SW`	No	`sw-KE-ZuriNeural`	Swahili TTS voice
`AZURE_TTS_VOICE_EN`	No	`en-US-JennyNeural`	English TTS voice
`AZURE_TTS_FORMAT`	No	`audio-16khz-128kbitrate-mono-mp3`	Audio format
`MAX_CALLS_PER_TICK`	No	`3`	Max outbound calls per scheduler tick
`CALL_SPACING_SEC`	No	`0.8`	Delay between outbound calls
`FLASK_SECRET_KEY`	No	`CHANGE_ME_NOW`	Flask session signing key
`AUTO_START_NGROK`	No	`1`	Auto-start ngrok on launch

Variable	Required	Default	Description
`AUTH_MAX_FAILS`	No	`7`	Failed attempts before lockout
`AUTH_LOCK_SECONDS`	No	`900`	Lockout duration (15 minutes)
`AUTH_WINDOW_SECONDS`	No	`600`	Failure tracking window (10 minutes)

9.2 Configuration File (`config.yaml`)

```yaml
twilio:
  account_sid: ""              # Overridden by .env
  auth_token: ""
  from_number: ""
  public_base_url: ""

ivr:
  questions_file: "data/questions.txt"
  gather_timeout_sec: 6
  speech_timeout: "auto"
  speech_language: "sw-KE"
  recording_max_seconds: 1800  # 30 minutes

audio_processing:
  enabled: true
  backend: "deepfilternet"
  processed_dir: "data/audio_processed"
  temp_dir: "data/audio_processed/tmp"
  output_sample_rate: 16000    # Whisper expects 16kHz
  model_sample_rate: 48000     # DeepFilterNet expects 48kHz
  channel_mode: "mixdown"      # "mixdown" or "channel"
  caller_channel: 0
  keep_intermediate_files: false

auth:
  users:
    username:
      password_hash: "pbkdf2:sha256:..."
```

9.3 System Constants

Constant	Value	File	Description
`MAX_ATTEMPTS`	`3`	`state.py`	Maximum call attempts per participant
`RETRY_GAP`	`1 hour`	`state.py`	Minimum time between retries
`RECORDING_MAX_SEC`	`1800`	`twilio_handler.py`	Max recording length
`GATHER_TIMEOUT`	`6s`	`config.yaml`	DTMF/speech input timeout
`SCHEDULER_INTERVAL`	`15s`	`twilio_handler.py`	Scheduler polling interval
`WORKER_POLL`	`5s`	`background_worker.py`	Worker polling interval
`MAX_CHARS`	`3000`	`translate.py`	Max characters per translation chunk
`SCHEDULER_TIMEOUT_SEC`	`45s`	`runtime_status.py`	Scheduler heartbeat timeout
`WORKER_TIMEOUT_SEC`	`20s`	`runtime_status.py`	Worker heartbeat timeout

---

10. Admin Dashboard

10.1 UI Layout

```
+------------------------------------------------------------------------+
| AudioSurvey AI — Admin                        [RUNNING] [PAUSED]       |
| NYC time: 2026-04-05 14:32:15 EDT     Logged in as: krishnanand        |
|                                                         [Sign out]     |
+------------------------------------------------------------------------+
| [Dial Now] [Start] [Stop] [Refresh] [Export Excel] [English] [Reset]  |
+------------------------------------------------------------------------+
| TECH INFO                                                              |
| Scheduler: Running (3s ago) | Worker: Running (1s ago) | Load: normal |
+------------------------------------------------------------------------+
| +------------+ +------------+ +------------+ +------------+           |
| |    42      | |    10      | |     5      | |    25      |           |
| |   Total    | |  Pending   | |In Progress| | Completed |           |
| +------------+ +------------+ +------------+ +------------+           |
+------------------------------------------------------------------------+
| Upload Contacts [CSV] | Questions Editor [Save] | Conference Call [Go]|
+------------------------------------------------------------------------+
| Participants                                                           |
| +------+----------+----------+----+----------+------+--------+-----+ |
| | ID   | Phone    | Status   |Att.| Engaged  |Dir.  |Sched.  |Ctrl | |
| +------+----------+----------+----+----------+------+--------+-----+ |
| | P001 | +2****34 | Completed| 1  | Engaged  |In    |04-05   |     | |
| | P002 | +2****56 | Pending  | 0  | Not eng. |Out   |        |[Set]| |
| | P003 | +2****78 | InProg.  | 1  | Engaged  |In    |        |[End]| |
| +------+----------+----------+----+----------+------+--------+-----+ |
+------------------------------------------------------------------------+
| [End All Calls]                                                        |
+------------------------------------------------------------------------+
```

10.2 New Dashboard Features (v1.2.0)

Feature	Description
Live Tech Info Panel	Shows Scheduler status, Worker status, and system load level (quiet/normal/busy) — updated every poll cycle
Call Direction Column	Shows `Incoming` or `Outgoing` badge per participant based on call direction
End Call Button	Per-participant button (visible when call is `in_progress`) to immediately terminate that call via Twilio API
End All Calls Button	Global button to terminate all currently in-progress calls at once
Refreshing Overlay	Visual overlay during state reset operations to prevent user interaction during reset
Redesigned Login Page	Stronger branding, animated gradient, "Welcome" header, improved UX
Runtime Status API	`/admin/live_state` now includes `runtime` object with scheduler and worker heartbeat info

10.3 Live Polling Architecture

```
  BROWSER (JS — every 1 second)      FLASK SERVER        participants.json
       |                                 |                      |
       |-- Guard checks:                 |                      |
       |   Skip if Flatpickr open        |                      |
       |   Skip if input dirty           |                      |
```

```
        |   Skip if poll in flight        |                     |
        |                                 |                     |
        |-- GET /admin/live_state -------->|                    |
        |                                 |-- load_participants |
        |                                 |-- get_runtime_snapshot()
        |<-- {total, counts,              |                     |
        |     participants, runtime}      |                     |
        |                                 |                     |
        |-- Update KPI cards              |                     |
        |-- Re-render table rows          |                     |
        |-- Update Tech Info panel        |                     |
        |-- Re-initialize Flatpickr       |                     |
```

---

11. Background Services

11.1 Service Lifecycle

```
   MAIN THREAD              SCHEDULER THREAD         WORKER THREAD
        |                         |                        |
        |-- start_background_services()                    |
        |-- Thread(daemon).start -->|                      |
        |-- Thread(daemon).start ---|-------------------->|
        |-- app.run(port=5050)      |                      |
        |                           |                      |
        |    +===[Every 15 seconds]=====+                  |
        |    | mark_scheduler_heartbeat()|                 |
        |    | load_participants()       |                 |
        |    | Filter can_call()         |                 |
        |    | client.calls.create()     |                 |
        |    | mark_call_started()       |                 |
        |    | save_participants()       |                 |
        |    +===========================+                 |
        |                                                  |
        |                     +===[Every 5 seconds]========+
        |                     | mark_worker_heartbeat()    |
        |                     | load_participants()        |
        |                     | Find processing_status     |
        |                     |   = "pending"              |
        |                     | Run ML pipeline            |
        |                     | mark_completed()           |
        |                     | save_participants()        |
        |                     +============================+
```

11.2 Worker Pipeline Stages

Stage	Progress	Typical Duration	Description
1. Prepare	10–25%	~5s	FFmpeg: extract channel, resample to 48kHz mono PCM16
2. Denoise	25–60%	~30–90s	DeepFilterNet: neural noise removal (PyTorch)
3. Resample	60–75%	~3s	FFmpeg: downsample to 16kHz for Whisper
4. Transcribe	75–85%	~60–180s	Whisper large-v3: Swahili STT
5. Translate	85–93%	~5–20s	Google Translate: chunked sw→en with retries
6. English TTS	93–97%	~5–10s	gTTS: English audio synthesis
7. Complete	100%	—	Mark completed, save outputs to state

11.3 Runtime Status Module (`runtime_status.py`)

Added in v1.2.0. Tracks heartbeats from both background threads and provides health status for the dashboard.

```
# Statuses returned by get_runtime_snapshot():
# "not_started"  -> Thread hasn't started yet
# "starting"     -> Thread started but no heartbeat yet
# "running"      -> Last heartbeat within timeout window
# "paused"       -> Scheduler is paused (admin toggled)
# "down"         -> No heartbeat received within timeout
```

---

12. Inbound Call Handling

12.1 Overview (v1.2.0)

Inbound calls arrive when a participant dials the Twilio phone number directly. The system routes them through the exact same IVR survey and ML processing flow as outbound calls.

12.2 Inbound Participant Matching

```
@app.route("/voice", methods=["POST"])
def voice():
    call_sid = request.values.get("CallSid")
    direction = request.values.get("Direction", "")  # "inbound" or "outbound-api"
    caller_number = request.values.get("From", "")   # E.164 phone number

    state = load_participants()
    pid, p = find_participant_by_callsid(state, call_sid)

    if not pid and direction == "inbound" and caller_number:
        # Try to match by phone number
        for existing_pid, existing_p in state.items():
            if existing_p.get("phone_e164") == caller_number:
                pid = existing_pid
                p = existing_p
                break

        if not pid:
            # Auto-create a new participant from the inbound number
            pid = f"inbound_{caller_number.replace('+', '').replace(' ', '_')}"
            upsert_participant(state, pid, caller_number)
            p = state[pid]

    # Tag direction on the participant record
    if pid:
        state[pid]["direction"] = direction
        mark_call_started(state, pid, call_sid)
        save_participants(state)
```

12.3 Call Controls (v1.2.0)

End Individual Call:

```
@app.route("/admin/end_call", methods=["POST"])
def end_call():
    call_sid = request.form.get("call_sid")
```

```
    client = Client(TWILIO_SID, TWILIO_TOKEN)
    client.calls(call_sid).update(status="completed")
```

End All Active Calls:

```
@app.route("/admin/end_all_calls", methods=["POST"])
def end_all_calls():
    state = load_participants()
    client = Client(TWILIO_SID, TWILIO_TOKEN)
    for pid, p in state.items():
        if p.get("status") == "in_progress" and p.get("last_call_sid"):
            client.calls(p["last_call_sid"]).update(status="completed")
```

---

13. Error Handling & Resilience

13.1 Fault Tolerance Matrix

Component	Failure Mode	Handling Strategy
Twilio Call	No answer / Busy	Mark `pending`, retry after 1h (up to 3 attempts)
Twilio Call	Failed / Canceled	Mark `failed` if max attempts reached
Azure TTS	API error	RuntimeError raised; corrupted audio not served
Azure TTS	Duplicate request	Disk cache by SHA1 hash prevents re-synthesis
Whisper	Hallucination drift	`condition_on_previous_text=False`, `temperature=0.0`
Google Translate	API error	3 retries with 1.5s backoff; failed chunks marked `[TRANSLATION_FAILED_CHUNK]`
Google Translate	Returns `None`	Explicit `None` check; RuntimeError raised
DeepFilterNet	Processing error	Graceful fallback to unprocessed audio
Recording Download	HTTP error	Logged; returns 200 (Twilio won't retry webhook)
State File	Corrupt JSON	Renamed to `.corrupt`, returns empty state
State File	Concurrent access	`threading.RLock()` + atomic write (`.tmp` then `os.replace`)
Scheduler	Exception in tick	Caught and logged; loop continues
Worker	Exception in pipeline	Caught; `processing_status="failed"`; loop continues
Auth State	Missing file	Returns safe defaults `{"fails": {}, "locks": {}}`
Inbound Call	Unknown caller	Auto-creates participant record from phone number
End Call API	Call already ended	Twilio returns 200; exception caught and logged

13.2 Retry Policy

```
Call Placed (attempt N of 3)
     |
     +-- completed + engaged ---------> Status: completed
     |                                   Pipeline: triggered
     |
```

```
        +-- completed + NOT engaged -----> Status: pending
        |                                   (retry in 1h if < 3 attempts)
        |
        +-- no-answer / busy / failed / canceled
                                |
                           attempts >= 3?
                                |          |
                               Yes         No
                                |          |
                           Status:     Status: pending
                           failed      (retry in 1h)
```

---

14. Logging & Observability

14.1 Logging Architecture

Log Type	Format	Location	Purpose
Application Log	Colored console (colorlog)	stdout	Real-time operational visibility
Auth Event Log	JSONL	`data/auth_log.jsonl`	Security audit trail
Call Log	CSV	`data/state/call_log.csv`	Call history, recording tracking, direction
Worker Progress	Terminal progress bar	stdout	ML pipeline progress
Runtime Status	In-memory heartbeats	`runtime_status.py`	Scheduler/worker liveness

14.2 Console Log Format

```
[NYC 2026-04-05T14:32:15-04:00 | UTC 2026-04-05T18:32:15Z] PROMPT SENT | CallSid=CA123 |
Participant=P001 | q3_mcq | Text="..."
[BackgroundWorker] participant=P001 | [################----]  80% | Removing background
noise
[Scheduler] 2026-04-05 14:32:30 EDT Calling P001 -> +254... | CallSid=CA123
```

14.3 Auth Event Log Format

```
{"event":"login","user":"krishnanand","ip":"127.0.0.1","login_utc":"2026-04-
05T18:32:15Z","login_local":"2026-04-05 14:32:15 EDT","user_agent":"Mozilla/5.0..."}
{"event":"logout","user":"krishnanand","session_duration_sec":13365}
```

14.4 Silenced Libraries

`httpx`, `httpcore`, `httpcore.connection`, `httpcore.http2`, `hpack`, `h2`, `gtts`, `gtts.tts`, `googletrans`, `urllib3`, `torio`, `torchaudio`

---

15. Deployment Guide

15.1 Prerequisites

Requirement	Version	Purpose
Python	3.10+	Runtime
pip	—	Package management
ngrok	3.x	HTTPS tunneling
FFmpeg	4.x+	Audio processing
Twilio Account	—	Voice API
Azure Account	—	Cognitive Services TTS

15.2 Local Development Setup

```bash
# 1. Clone and enter the repository
git clone https://github.com/krishnanand20/audiosurvey_ai.git
cd audiosurvey_ai

# 2. Create a virtual environment
python3 -m venv venv
source venv/bin/activate

# 3. Install all dependencies
pip install -r requirements.txt

# 4. Configure your credentials
cp .env.example .env
# Edit .env with your Twilio + Azure keys

# 5. Launch (auto-starts ngrok + Flask + opens admin in browser)
python3 run_app.py
```

15.3 Startup Sequence

```
python3 run_app.py
      |
      +-- Check AUTO_START_NGROK env
      +-- Check if ngrok already running (localhost:4040)
      |
      +-- [No existing tunnel] Start ngrok http 5050
      +-- Poll /api/tunnels until HTTPS URL appears (20 retries)
      +-- Set PUBLIC_BASE_URL env var
      |
      +-- subprocess: python3 -m app.twilio_handler serve
          |
          +-- load_dotenv() + load_config()
          +-- Validate Twilio env vars
          +-- Load Whisper model large-v3 (cached globally)
          +-- init Azure TTS SDK
          +-- start_background_services()
          |    +-- Scheduler thread (daemon, 15s loop)
          |    +-- Worker thread (daemon, 5s poll)
          |
          +-- app.run(host="0.0.0.0", port=5050)
      |
      +-- Wait 1.5s
      +-- webbrowser.open("http://127.0.0.1:5050/admin")
```

15.4 macOS DMG Distribution

```
cd packaging/macos_dmg
./build_macos_dmg.sh
# Output: output/AudioSurvey-AI.dmg
```

Build steps: Swift icon generation → `.iconset` → `.icns` → `.app` bundle with launcher script → DMG

---

16. Directory Structure

```
audiosurvey_ai/
|
|-- .env                         # Environment variables (secrets — gitignored)
|-- .gitignore                   # Git exclusion rules
|-- config.yaml                  # Application configuration
|-- main.py                      # Batch processing entry point
|-- run_app.py                   # Application launcher
|-- requirements.txt             # Python dependencies (146 packages)
|-- README.md                    # Quick-start
|-- DOCUMENTATION.md             # This file
|-- STUDY_GUIDE.md               # Presentation study guide
|
|-- app/
|   |-- twilio_handler.py        # Flask app, IVR routes, inbound handling, auth (~1,780
lines)
|   |-- dashboard.py             # Admin dashboard UI + routes (~1,280 lines)
|   |-- runtime_status.py        # Background service health tracking (NEW v1.2.0)
|   |-- state.py                 # Thread-safe state management
|   |-- scheduler.py             # Background call scheduler
|   |-- background_worker.py     # ML processing pipeline worker
|   |-- audio_preprocess.py      # DeepFilterNet noise removal
|   |-- transcribe.py            # Whisper speech-to-text
|   |-- translate.py             # Google Translate integration
|   |-- tts.py                   # Google TTS (English audio)
|   |-- azure_tts.py             # Azure Cognitive Services TTS
|   |-- export_excel.py          # Excel response export
|   |-- auth.py                  # Authentication helpers
|   |-- utils.py                 # Scheduling utilities
|   |-- file_naming.py           # Safe filename generation
|   |-- logger.py                # Colored logging setup
|   |-- runtime_warnings.py      # Warning suppression
|   |-- twilio_utils.py          # Twilio call helpers
|
|-- data/
|   |-- questions.txt            # Survey questions (pipe-delimited)
|   |-- state/
|   |   |-- participants.json    # All participant records, responses, status
|   |   |-- call_log.csv         # Call history with direction field
|   |   |-- settings.json        # System settings (paused flag)
|   |-- audio/                   # Raw call recordings (.wav)
|   |-- audio_processed/         # Denoised recordings (.wav)
|   |-- transcripts/             # Whisper transcriptions (.txt, Kiswahili)
|   |-- translations/            # English translations (.txt)
|   |-- english_audio/           # English TTS audio (.mp3)
|   |-- ivr_audio/               # Cached IVR prompts (.mp3, hash-named)
|   |-- results/
|       |-- ivr_responses.xlsx    # MCQ/MCQO responses (original)
|       |-- ivr_responses_english.xlsx # MCQ/MCQO responses (English)
```

```
  |
  |-- packaging/
      |-- macos_icon_generator.swift
      |-- macos_dmg/
          |-- build_macos_dmg.sh
          |-- output/AudioSurvey-AI.dmg
```

---

17. Dependency Matrix

17.1 Core Dependencies

Package	Version	Purpose	Critical?
`Flask`	3.1.2	Web framework	Yes
`twilio`	9.10.0	Telephony API (calls, TwiML, webhooks)	Yes
`openai-whisper`	20240930	Post-call speech-to-text (large-v3)	Yes
`azure-cognitiveservices-speech`	1.48.1	Neural TTS for IVR prompts	Yes
`googletrans`	4.0.0rc1	Kiswahili → English translation	Yes
`gTTS`	2.5.4	English audio generation	Yes
`deepfilternet`	0.5.6	Background noise removal	Medium
`torch`	2.8.0	ML framework (Whisper + DeepFilterNet)	Yes
`torchaudio`	2.8.0	Audio tensor operations	Yes

17.2 Data & Export

Package	Version	Purpose
`pandas`	2.3.3	DataFrame operations for Excel export
`openpyxl`	3.1.5	Excel (.xlsx) file writing
`numpy`	1.26.4	Audio data numerical operations
`pydub`	0.25.1	Audio manipulation

17.3 Infrastructure

Package	Version	Purpose
`python-dotenv`	1.2.1	`.env` file loading
`PyYAML`	6.0.3	`config.yaml` parsing
`gunicorn`	23.0.0	Production WSGI server
`Werkzeug`	3.1.4	Password hashing, HTTP utilities
`colorlog`	6.10.1	Colored console logging
`requests`	2.32.5	HTTP client (recording download)
`Jinja2`	3.1.6	Template rendering

17.4 External Services

Service	Provider	Purpose	Pricing

Service	Provider	Purpose	Pricing
Voice API	Twilio	Outbound/inbound calls, recording, webhooks	Per-minute
Speech TTS	Azure Cognitive Services	Neural TTS prompts (Swahili + English)	Per-character
Translation	Google Translate (googletrans)	Kiswahili → English	Free (scraping)
Text-to-Speech	Google TTS (gTTS)	English audio output	Free
Tunneling	ngrok	HTTPS tunnel for Twilio webhooks	Free tier

---

18. Glossary

Term	Definition
IVR	Interactive Voice Response — automated phone menu system
DTMF	Dual-Tone Multi-Frequency — phone keypad tones (pressing 1–9, *, #)
TwiML	Twilio Markup Language — XML instructions for call scripting
TTS	Text-to-Speech — converting text to spoken audio
STT	Speech-to-Text — converting spoken audio to text
SSML	Speech Synthesis Markup Language — XML for controlling TTS prosody and voice
E.164	International phone number format (e.g., `+254700000000`)
CallSid	Unique Twilio identifier for a phone call
Direction	Call direction: `inbound` (participant called us) or `outbound-api` (we called them)
DeepFilterNet	PyTorch neural network for real-time speech enhancement and noise removal
Whisper	OpenAI's multilingual speech recognition model (large-v3 used here)
MCQ	Multiple Choice Question — DTMF keypad input
MCQO	Multiple Choice with Other — DTMF + optional speech for "Other" option
OPEN	Open-ended question — free speech input
INFO	Informational prompt — no response collected
Engaged	Participant produced detectable real speech during the call
Pipeline	The 4-stage ML chain: DeepFilterNet → Whisper → Google Translate → gTTS
Heartbeat	Periodic signal emitted by background threads to indicate they are alive
Atomic Write	Write to temp file then `os.replace()` — prevents partial/corrupted file writes
RLock	Reentrant Lock — a thread lock the same thread can acquire multiple times
Daemon Thread	Background thread that automatically exits when the main process ends
ngrok	Tunneling service that exposes localhost to the internet over HTTPS
Kiswahili / sw	Swahili language (ISO 639-1: `sw`; BCP 47 locale: `sw-KE`)
PBKDF2	Password-Based Key Derivation Function 2 — slow hashing for password storage
PII	Personally Identifiable Information (phone numbers, names) — kept gitignored
Flatpickr	JavaScript date/time picker library used in the admin dashboard

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